Exercise Sheet 3

The spaces are complex and finite-dimensional, except in Exercise 3.6. The matrix unit E_{ij} has entry 1 in position (i, j) and zero elsewhere. Recall the rule $E_{ij}E_{kl} = \delta_{jk}E_{il}$.

The closure lemma of Lecture 3 states that solvability passes to subalgebras and homomorphic images. It also passes from an ideal and the quotient by it to the whole algebra.

Exercise 3.1 (Series of $\mathfrak{b}_3$). Let $\mathfrak{b}_3$ be the Lie algebra of upper triangular 3×3 matrices, and $\mathfrak{n}_3$ its subalgebra of strictly upper triangular matrices.

(i) Compute the derived series of $\mathfrak{b}_3$.

(ii) Compute its lower central series. Is $\mathfrak{b}_3$ nilpotent?

(iii) Find a common eigenvector of $\mathfrak{b}_3$ in $\mathbb{C}^3$ and its eigenvalue function. Show that it spans the only invariant line.

Exercise 3.2 (Solvability through flags). Let $\dim \mathfrak{g} = n$.

(i) Prove that $\mathfrak{g}$ is solvable if and only if $\operatorname{ad}\mathfrak{g}$ is solvable.

(ii) Prove that $\mathfrak{g}$ is solvable if and only if there are ideals I_k of $\mathfrak{g}$ with

$$0 = I_0 \subset I_1 \subset \cdots \subset I_n = \mathfrak{g}, \qquad \dim I_k = k \quad (0 \le k \le n).$$

(iii) Let $\mathfrak{g}$ be solvable. Prove that $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.

(iv) Prove conversely that $\mathfrak{g}$ is solvable if $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.

Exercise 3.3 (Two-dimensional modules of $[h, e] = e$). Let $\mathfrak{g} = \mathbb{C}h \oplus \mathbb{C}e$ with $[h, e] = e$.

(i) Prove that $\rho(e)$ is nilpotent in every module.

(ii) Let $\dim V = 2$ and $\rho(e) \neq 0$. Find a basis and $a \in \mathbb{C}$ with

$$D(h) = \begin{pmatrix} a+1 & 0 \\ 0 & a \end{pmatrix}, \qquad D(e) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

(iii) Show that different values of a give nonisomorphic modules, and that each has exactly one invariant line.

Exercise 3.4 (Radicals).

(i) For Lie algebras $\mathfrak{g}_1, \mathfrak{g}_2$, prove $\operatorname{rad}(\mathfrak{g}_1 \oplus \mathfrak{g}_2) = \operatorname{rad}\mathfrak{g}_1 \oplus \operatorname{rad}\mathfrak{g}_2$.

(ii) Prove $\operatorname{rad}\mathfrak{gl}_2(\mathbb{C}) = \mathbb{C}I_2$ and $\mathfrak{gl}_2(\mathbb{C})/\mathbb{C}I_2 \cong \mathfrak{sl}_2$.

(iii) The upper triangular algebra $\mathfrak{b}_2$ is solvable but is not contained in $\mathbb{C}I_2$. Why does this not contradict (ii)?

Exercise 3.5 (Converse statements that fail).

(i) Show that for each $X \in \mathfrak{sl}_2$ some basis of $\mathbb{C}^2$ makes the matrix of X upper triangular, but no basis does this for all elements at once.

(ii) Let $\mathfrak{g} = \mathfrak{sl}_2 \oplus \mathbb{C}t$, with t central, act on $\mathbb{C}$ by $\rho(X + ct) = c$. Show that ρ is a nonzero triangular representation and $\mathfrak{g}$ is not solvable. The triangular-representation theorem says that a faithful representation is triangular exactly when the algebra is solvable. Which hypothesis of this statement fails?

(iii) Find an ideal $\mathfrak{a}$ of $\mathfrak{b}_2$ such that $\mathfrak{a}$ and $\mathfrak{b}_2/\mathfrak{a}$ are nilpotent, while $\mathfrak{b}_2$ is not.

Exercise 3.6 (The hypotheses of Lie's theorem). This exercise departs from the standing field $\mathbb{C}$.

(i) Over $\mathbb{R}$, let $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ span a Lie algebra acting on $\mathbb{R}^2$. Show that it is solvable and has no common real eigenvector. What happens over $\mathbb{C}$?

(ii) Let p be prime and let $V = \mathbb{F}_p^p$ have basis $v_0, \dots, v_{p-1}$, indices mod p. Define $xv_i = v_{i+1}$ and $yv_i = iv_i$. Show that $\operatorname{span}\{x, y\}$ is a two-dimensional solvable Lie algebra with no common eigenvector. Locate the failure in the proof of the invariance lemma.

Exercise 3.7 (Jordan parts and triangular form). Recall the linear Jordan decomposition $T = T_s + T_n$ from the lecture.

(i) Let $T = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$. Find T_s and T_n. Is T_s the diagonal part of T?

(ii) Let the one-dimensional algebra $\mathbb{C}t$ act on $\mathbb{C}^2$ so that $\rho(t)$ has the upper triangular matrix $D(t) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ in the standard basis. Find $D(t)_s$ and $D(t)_n$, and show that neither lies in $\mathbb{C}D(t)$.